Question 1

The value of a real estate investment is determined by an assessment of its risks and opportunities.

a- What are the main risks involved in investing in real estate? How do they differ from the risks that characterise investment in other financial assets? (10 points)

These issues were discussed at length during the first part of Lecture 1 and again in Lecture 3. Outstanding students should also link this discussion to the lecture by Lois Liao on social and environmental perspectives in real estate finance – as she talked extensively about risk and opportunities.

b- 'Indivisibility, large-lot size and illiquidity'. Discuss these three terms in relation to real estate investment, and discuss how they affect an asset's price and riskiness (10 points)

These issues were also discussed in Lecture 1 and again in Lecture 3 and 4, when leverage was introduced. Amongst other things, students should emphasise that these issues imply leverage is an intrinsic part of RE investment (either because of lack or equity or because of portfolio considerations). This has important implications for risk and pricing.

c- What are the main opportunities from investing in real estate? (5 points)

These issues were discussed in Lecture 1. It would be good if students emphasised the fact that rental values are more "important" than sale prices.

d- Discuss how cap rates (or exit yields) can be used to estimate the resale value of a real estate asset. How do you adjust (up or down) prevailing cap rates in order to take into account uncertainty in economic conditions at the time of sale? (5 points)

Lecture 2 and 3 discussed these points (exit yields or cap rates).

e- Cap rates are said to be myopic. What does this mean? Articulate your answer by constructing an example based on the following instructions: *a*- consider an investment horizon of 5 years; *b*- consider two properties – one with stable Net Operating Income (NOI) at 1mio (property A) and one with initial NOI at 1mio but growing at 5% per year (property B); *c*- consider an unadjusted cap rate of 10%. Using this information, determine the NOIs for both properties in each of the 5 years and the sale values at the end of the holding period. How much would you pay for Property A and Property B if you worked out your answer using an uncorrected cap-rate method? How would this answer change if instead you used a Discounted Cash Flow (DCF) method using discount factor equal to 10% (20 points)

The example can be constructed as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Prop A – NOI	1	1	1	1	1	Not needed
Prop A – Sale Price					10 (1/10%)	
Prop A – NOI	1	1.05	1.1025	1.1576	1.2155	1.2763
Prop A – Sale Price					25.5260 (1.2763/{10%-5%})	

Using a cap-rate both properties would be worth 10mio at Year 0, but this clearly cannot be the case as one property is returning a higher stream of future NOIs and sale price. Using a DCF, the answers are 10mio and 20mio, which is correct given streams of future returns.

f- Discuss the findings at point e-, and clarify how a 'Gordon Model' corrected cap-rate addresses the issues that emerge from this example. Provide both an intuitive answer and a back-of-the-envelope calculation showing how a corrected-cap rate solves inconsistences between a simple cap-rate method and the DCF approach (10 points)

If the students use a corrected cap-rate, they would find 10 and 20 as underpinned by the DCF. This is because the corrected cap-rate allows to take into account growth – and incorporate in one number both 'risks and opportunities'. These issues were discussed at length in Lectures 2 and 3.

g- Real Estate Investment Trusts (REITs) allow investors to by-pass some of the downsides of real estate, while capturing some of its benefits. Discuss this statement clarifying: *a*- the main features that characterise REITs, including benefits, constraints and growth strategies; *b*- how their performance tracks or predicts returns in direct real estate investment; and *c*- the main determinants of their relative performance and value fluctuations over the business-cycle (20 points)

These points were discussed several times during Lectures 8 by the regular lecturer and by the external speaker.

h- Discuss the most important issues to bear in mind when structuring a portfolio of indirect real estate investments. How would you take into account 'social and

environmental' factors when thinking about the performance of your assets? And how would you take into account UK's 'Net Zero Carbon Target' when structuring your portfolio? (20 points)

This question combines the insights provided by the guest speakers Lois Liao and Niko Szumilo, respectively in Lectures 7 and 9.

Question 2

You are considering investing in a property worth £30,000,000 and hold it for five years. The initial monthly Net Operating Income (NOI) is estimated to be £200,000 and grow at 5% per year. You are discussing with the bank the following lending facility: a constant payment mortgage (CPM) with a Loan-To-Value (LTV) ratio of 80%, monthly compounding, stretching over 20 years and with an annual interest rate of 7.5% per year.

a- What is the value of the constant monthly payments on this loan? (10 points)

Loan structures and all the material to answer this question (all points) were discussed in Lectures 4 and 5. The answer to this specific point is £193,342.3665.

b- You are planning to settle the debt when you sell the property at the end of Year 5. What is the outstanding principal at that point? How much interest have you paid up to that point? How much have you paid in amortization? (5 points)

These numbers can easily be found using the AMROT worksheet of the calculator and they are respectively: £20,856,503.62; £8,457,045.61; and £3,143,496.380.

c- What is the effective cost of borrowing of this loan? Comment on your answer (5 points)

The effective cost of borrowing is identical to the nominal cost of borrowing. No need to carry out calculations. This comes straight from theory: when there are no origination charges and penalties, nominal and effective costs are the same.

d- Consider now an Interest Only (IO) loan as an alternative. It also stretches over 5 years, has the same LTV and interest rate, it is repaid after 5 years and has monthly compounding. What is the monthly payment of such loan? And how much is the final balloon payment worth? Do you prefer this loan or the CPM mortgage at point a-? And which loan do you qualify for if the bank uses a Debt-Coverage Ratio (DCR) of 1.2? (10 points)

The monthly payments of this loan are 150,000 (remember the monthly interest is 7.5%/12). The balloon payment is 24,150,000 (interest and principal reimbursement). The IO and the CPM have the same effective cost of borrowing. With a DCR of 1.2 the borrower only qualifies for the IO loan.

e- Revert now to the original CPM mortgage. The lender includes five discount points. You still repay the loan when you sell the property after 5 years. What is the effective cost of borrowing? And how does this change if the lender further imposes a 5% penalty charge on the outstanding loan at the time of reimbursement? (15 points)

Students need to keep in mind out the outstanding loan at the end of year five worked out above (£20,856,503.62). They then need to recognise that with 5 discount points they only receive 22,800,000 but still pay 193,342.3665 per month. They then need to work out the IRR of this cash flow, which is 8.8131% without pre-payment penalty or 9.5566% if the 5% fee is also charged on the outstanding balance.

f- What would be the effective cost of borrowing (with discount points and penalty charges) if you repaid after three years? And what if you paid after 1.5 years? Comment on your findings – what do you notice? (10 points)

The balance after three years is 22,256,276.71 giving an effective cost of borrowing of 10.88%. After 1.5 years (18months), the balance is 23,176,976.77, giving an effective cost of borrowing of 14.2308%. The cost is larger, the sooner the borrower repays. This is because discount points are a sunk cost that is spread over a larger/smaller number of months depending on prepayment behaviour – affecting the percentage costs of the loans.

g- What is the rationale for banks to impose discount points? And why do some borrowers have incentives to accept loans with discount points? (10 points)

These issues were discussed in Lectures 4 and 5.

h- The bank is now offering an alternative loan structured as follows. The LTV is still at 80% and the nominal annual interest rate is 7.5%. The compounding is monthly and the length of mortgage is 20 years. However, the loan is not CPM, but a Graduated Payment Mortgage (GPM) with a 5% graduation rate. This graduation rate applies between Year 1 and 2, between Year 2 and 3 and finally between Year 3 and Year 4. After that, the value of the monthly payments stabilises. What are the payment values in months 1, 13, 25 and 37? (20 points)

Similar calculations to these were discussed during Lecture 5. Essentially students need to realise that this is a bundled discount cash flow calculation with the initial payment unknown. This can be structured as follows:

$$24mio = PMT\sum_{n=1}^{12}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}} + (1+5\%)PMT\sum_{n=1}^{12}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{12}}$$

$$+ (1+5\%)^{2}PMT\sum_{n=1}^{12}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{24}}$$

$$+ (1+5\%)^{3}PMT\sum_{n=1}^{204}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}}\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{36}}$$

The following terms can be calculated as discounting factors or present values of annuities:

$$\sum_{n=1}^{12} \frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}} = 11.5264$$

$$\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{12}} = 0.9280$$

$$\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{24}} = 0.8611$$

$$\frac{1}{\left(1+\frac{7.5\%}{12}\right)^{36}} = 0.7991$$

$$\sum_{n=1}^{204} \frac{1}{\left(1+\frac{7.5\%}{12}\right)^{n}} = 115.1133$$

The other expressions can also be easily calculated to give the following:

24mio= PMT * 11.5264 + PMT * 1.05 * 11.5264 * 0.9280 + PMT * 1.1025 * 11.5264 * 0.8611 + PMT * 1.1576 * 115.1133 * 0.7911

Summing up, this gives 24mio = PMT * 140.1846, giving PMT = 171,202.8283. This then goes up at yearly intervals till the beginning of year 4, as follows: 179,762.9697; 188,751.1182; and 198,188.6741. If the students use these payments to do a back of the envelope calculations, they should find an IRR of 7.5%.

i- Imagine now you are repaying this GPM loan after 1.5 years. The bank is still charging 5 discount points and a pre-payment penalty fee of 5% on the outstanding balance of the mortgage. What is the effective cost of borrowing of this operation? Compare this to the effective cost of borrowing if repaying after 1.5 years at point f-; what do you notice? (15 points)

In order to answer this question, students have to work out the outstanding balance of the loan after 18months. This has to be done “manually”, rather than using the amortization worksheet. To begin with, students have to consider that they have taken a 24mio loan charging 7.5% but are paying less than the equivalent CPM loan. So they have to compute the future value of a loan that is not fully amortizing. To do so, they have to use 24mio as PV, 7.5% as I/Y, 12 as N and then 171,202.8283 as PMT for the first 12months. This gives an FV of 23,736,635.09. This is now the new PV for a calculation stretching for the next 6months with PMT of 179,762.9697 and I/Y of 7.5% and N of 6. This gives an FV of 23,545,211.84. This is what is

required to calculate the effective cost of borrowing. This calculation has 22,800,000 as initial loan value (after deducting discount points); 12 months at 171,202.8283; 5 months at 179,762.9697; and 1 month 179,762.9697 plus the outstanding balance 23,545,211.84, further increased by the 5% penalty charge. This gives a 14.2263% - smaller than the IRR at point f-. The students should provide a rationale for this finding based on the actual size of the payments of the GPM relative to the CPM.

Question 3

You are planning to buy a shopping centre with an investment horizon of five years. All the available space is distributed on one floor and the total rentable area consists of 10,000 square metres. Of these, 5,000 square metres are rented to a supermarket, which uses 4,000 metres for the shopping and retail area and the remaining 1,000 for stocks and supplies. The rents per square metre per year are £350 for the shopping/retail portion and £250 for the stocks/supplies area. A lease with the supermarket (starting in Year 1) has already been signed. At the end of Year 3 there will be rental review and you expect rents to increase by 10% (and stabilise thereafter). The remaining 5,000 metres are evenly divided among five other shopping spaces. At the moment, these are empty. You expect to refurbish these spaces in the first 6 months of your holding period at the cost of £100 per square metre; then you expect the shops to remain vacant for 6 months and to spend £50 per square metre on marketing and brokering to let the spaces to some tenants. To simplify matters, assume that both the refurbishment costs and the marketing costs are paid at the end of Year 1. You then expect the five shops to be let to five tenants and to have identical contracts with the following features: *a*- The rent per square metre per year will be £380; *b*- You will give your tenants 12months rent-free concessions; *c*- The rents will be reviewed three years after the contracts are signed; and *d*- At review, you expect rents to increase by 10% (and stabilise thereafter). While the space is being refurbished, you expect to pay the supermarket a compensation of £500,000 – also to be paid at the end of the year in which the disruption occurs. Finally, you estimate that the prevailing cap-rates (yields) in the area to be 10%.

a- Using this information, put together the pro-forma that describes the operations and resale value of this investment. (20 points)

The pro-forma is as follows (note that it extends to year 6 to calculate the resale value; however as nothing changes between Year 5 and Year 6 this is relative redundant – but it would be good for students to explain that what they need is the NOI the year after sale):

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Supermarket	4000m *£350 + 1000m *£250 -£500,000 compensation for Disruption £1,150,000	4000m *£350 + 1000m *£250 £1,650,000	4000m *£350 + 1000m *£250 £1,650,000	4000m *£350 * (1+10%) + 1000m *£250 * (1+10%) = 4000m *£385 + 1000m *£275 £1,815,000	4000m *£385 + 1000m *£275 £1,815,000	4000m *£385 + 1000m *£275 £1,815,000
Other shops (no need to calculate each shop separately as they all share the same contracts)	-£100 * 5000m (refurbishment) - £50 * 5000m (advertising) = -£750,000	5000m * £380 = £1,900,000 Set to 0 because of rent free concession £0	5000m * £380 = £1,900,000	5000m * £380 = £1,900,000	5000m * £380 * (1+10%) = 5000m * £418 * £2,090,000	5000m * £380 * (1+10%) = 5000m * £418 * £2,090,000
Grand total	£400,000	£1,650,000	£3,550,000	£3,715,000	£3,905,000	£3,905,000

With a cap-rate (yield) of 10% and based on the NOI of Year 6, this means a resale price of £39,050,000

b- You are now asked to pay £29,000,000 to buy this real estate asset. Given the pro-forma put together at point a-, what is the implied Internal Rate of Return (IRR)? (5 points)

Such calculations were performed several times both during the lectures and the classes. The students should be able to work this out quite easily. The IRR is 13.5897%.

c- You now decide to flex some of the assumptions of your cash flow projections. In particular, you consider the possibility that cap-rates might be two percentage points lower and the possibility that cap-rates might be two percentage points higher than what discussed at point a-. Considering that each of these three alternatives (including the central scenario described a point a-) occurs with a 1-in-3 likelihood, what is the expected IRR of your operation? And what is the standard deviation around the expected IRR? Discuss what you notice and explain what other elements of your cash flow you would flex to assess the 'robustness' of your returns. (10 points)

While this seems like a complicated question, it is not. Students simply have to adjust the sale price and re-calculate an IRR by changing only the data for the last year of the holding period. With a lower cap-rate (8% instead of 10%), the sale price is higher – at £48,812,500 – giving an IRR of 17.6845%. With a higher cap-rate (12% instead of 10%), the sale price is lower – at £32,541,666.67 – giving an IRR of 10.4659%. The average IRR is 1/3 * 13.5897% + 1/3 * 17.6845% + 1/3 * 10.4659% = 13.9%. To work out the standard deviation, students first need the variance: 1/3 * (13.5897% - 13.9%)^2+ 1/3 * (17.6845% - 13.9%)^2 + 1/3 * (10.4659% - 13.9%)^2 =0.087%; the standard deviation is the square root of the variance and it is equal to2.95%. The students should provide comments about the relative riskiness of this investment, and also discuss what other elements they would like to flex (rents, vacancies, rent-free periods) in order to assess how sensitive their investment is to other assumptions. There is no set-answer and students are not required to do more calculations – but these issues were discussed time and again during the first five lectures of the course and by the case study speakers from Tristan.

d- You now consider taking a loan from the bank. This would be an interest-only (IO) loan over five years, with an LTV of 25% and charging 5% per year. What is the IRR after leverage? And what would be the implied return-on-equity using a Weighted-Average Cost of Capital (WACC) approximation? Comment on your results contrasting what you obtain when computing a fully-fledged IRR to what you find when using a WACC formula. (15 points)

Again, this is a pretty straight-forward question. The data for the loan are as follows. LTV 25% of £29,000,000 = £7,250,000. This is the amount borrowed. Charged at 5% per year, this

means annual interest payments of £362,500. The loan is reimbursed in Year 5 when the property is sold. This gives an IRR of 15.9533%. The WACC formula would instead be as follows: R(e) = 13.5897% + (13.5897% - 5%) * 25%/75% (the last term is the debt-to-equity ratio). This gives a 16.4529%. This is slightly different from the IRR because the WACC is an approximation that does not take into account the fact that D/E does not stay constant during the holding period.

e- You are now offered a participation loan with the following characteristics. The mortgage would still be an interest-only (IO) loan over five years, with an LTV of 25% and charging 5% per year. However, the bank would receive 25% of any annual NOI in excess of £3,000,000 and 25% of any increase in the value of property at the time of sale. What is now the IRR of your investment? And what is the effective cost of borrowing [hint: this is the same as the IRR earned by the bank on this lending operation]? Discuss your findings (20 points)

A similar exercise was discussed during one of the classes of the course. The solution is not very difficult. The data for the amount of the loan and the yearly interest payments don't change. These are respectively 7,250,000 and 362,500. However, starting from Year 3 the bank receives 25% of the NOI above 3,000,000. The figures would be as follows:

- Year 3: (3,550,000 – 3,000,000) * 25% = 137,500
- Year 4: (3,715,000 – 3,000,000) * 25% = 178,750
- Year 5: (3,905,000 – 3,000,000) * 25% = 226,250

On top of these additional payments that are subtracted from the amounts earned by the equity holder, the bank received 25% of the asset appreciation. That is: (39,050,000 – 29,000,000) * 25% = 2,512,500. Putting numbers together, this results in an IRR for the investor of 14.1651%.

For the bank, the revenues are as follows:

Year 0 = -7,250,000

Year 1 = 362,500

Year 2 = 362,500

Year 3 = 362,500 + 137,500

Year 4 = 362,500 + 178,750

Year 5 = 7,250,000 + 362,500 + 226,500 + 2,512,500

This gives rise to an IRR of 11.7824% - which corresponds to the effective cost of borrowing.

I except students to discuss how the participation loan clearly reduces the profitability of the operation but provides the bank with additional revenues.

f- You are finally considering an alternative funding strategy: a private equity deal. You will act as the project manager and General Partner (GP) and will team up with a set of investors that will be Limited Partners (LP) in the deal. You have agreed that as a GP you will invest £9,000,000 and the LPs will invest £20,000,000. The LPs will receive a 5% "pref" on their equity which is non-cumulative. As the GP, you will also receive a 5% interest on the equity that you inject as part of the "money". Once these coupons have been paid, extra revenues from the operations will be split 50-50 between GP and LPs. It was also agreed that at sale the LPs will recover their capital and then you will recover your equity. The LPs have also been promised an overall return of 10%. Residual sale proceedings after this target has been met will be split 50-50. What is the IRR of this investment for the LPs? And for the GP? (20 points)

The solutions are as follows. The 5% coupon on the 20,000,000 equity of the LPs is worth 1,000,000, while the 5% coupon on the 9,000,000 equity of the GP is worth 450,000.

The overall operation cash flows are reported above (see point a-); the distribution of funds between GP and LPs is as follows

	LP		GP		Residual
Year 1	400,000 (all income to LPs)		--		--
Year 2	1,000,000 (coupon) + 100,000 (50% of residual)		450,000 (coupon) + 100,000 (50% of residual)		200,000
Year 3	1,000,000 (coupon) + 1,050,000 (50% of residual)		450,000 (coupon) + 1,050,000 (50% of residual)		2,100,000
Year 4	1,000,000 (coupon) +		450,000 (coupon) +		2,265,000

	1,132,500 (50% of residual)		1,132,500 (50% of residual)		
Year 5	1,000,000 (coupon) + 1,227,500 (50% of residual)		450,000 (coupon) + 1,227,500 (50% of residual)		2,455,000

The cash flow for the LPs is therefore as follows.

Year 0: -20,000,000

Year 1: 400,000

Year 2: 1,100,000

Year 3: 2,050,000

Year 4: 2,132,500

Year 5: 2,227,500 + 20,000,000

We now need to determine the additional amount from the sale that given to the LPs in Year 5 will make this cash flow earn 10%. This has to be done manually working through the following expressions:

$$20{,}000{,}000 = \frac{400{,}000}{(1+10\%)} + \frac{1{,}100{,}000}{(1+10\%)^2} + \frac{2{,}050{,}000}{(1+10\%)^3} + \frac{2{,}132{,}500}{(1+10\%)^4} + \frac{2{,}227{,}500\ +\ 20{,}000{,}000}{(1+10\%)^5} + \frac{\mathrm{x}}{(1+10\%)^5}$$

We need to work out x above. These numbers are as follows:

20,000,0000 = 363,636.36+909,090.905+1,540,195.342+1,456,526.194+1,383,102.247+12,418,426.46+x/1.6105, giving x=3,106,690.721. This extra gives an IRR of 10%.

After this has been paid out of the 39,050,000 at sale alongside the 20,000,000 for the LPs and 9,000,000 for the GP, the remaining funds can be distributed 50-50. This means 6,943,309.279 can be split into 3,471,654.639 for GP and same for LPs. The new cash flow for the LPs is therefore:

Year 0: -20,000,000

Year 1: 400,000

Year 2: 1,100,000

Year 3: 2,050,000

Year 4: 2,132,500

Year 5: 2,227,500 + 20,000,000 + 3,106,690.721 + 3,471,654.639

This has an IRR of 12.4767%.

For the GPs, the cash flow is as follows:

Year 0: -9,000,000

Year 1: 0

Year 2: 550,000

Year 3: 1,500,000

Year 4: 1,582,500

Year 5: 14,149,154.64

This gives an IRR of 15.9442%

g- Imagine now there is a "lookback clause" capping the returns for the LPs at 10% and distributing all the proceeds from the sale after this target return has been achieved to the GP. What is the IRR for the GP? And why would the LPs accept such a clause? (5 points)

These issues were discussed in Lecture 10. The calculations are simple once point f- has been worked out. We already know how much out of the sale value needs to be paid to the LPs to achieve 10%. All the rest simply goes to the GP and increases their IRR. The numbers are as follows:

Year 0: -9,000,000

Year 1: 0

Year 2: 550,000

Year 3: 1,500,000

Year 4: 1,582,500

Year 5: 450,000 + 1,227,500 + 9,000,000 + 6,943,309.279 (the latter figure was previously split 50-50, now it all goes to the GP).

This gives an IRR of 20.2731%.